PROGRESS PRESENTATION #2

ICU Length-of-Stay Prediction

From a finalized GRU baseline to text embeddings, full explainability, time-series monitoring analysis, and a first prototype-retrieval model.

DATASET	TASK	COHORT	TEST SET
MIMIC-IV · MIMIC-CXR	Binary: LOS > 7 days	30,615 ICU stays	4,593 patients · 23.6% positive

GRU Baseline · BioClinicalBERT · SHAP / TimeSHAP · Monitoring Analysis · PRD-Net Prototype

Agenda

OVERVIEW

1. **Finalize Baseline Model**
   Multi-branch GRU over 48h vitals + 177 static features — finalized test performance

2. **Including Textual Embeddings (Comparison)**
   BioClinicalBERT radiology embeddings — full cohort vs. CXR-only ablation

3. **Finalizing Explainability**
   SHAP & TimeSHAP — which features and which ICU hours drive predictions

4. **Time-Series Analysis (Frequency of Values)**
   How often each ICU variable is actually measured over the first 48h

5. **First Prototype Model (PRD-Net)**
   Peer-retrieval network — results compared against the baseline & literature

SECTION 01

Finalize Baseline Model

A multi-branch GRU over 48h of vitals and 177 static clinical features — the reference model for everything that follows.

01

The Clinical Problem & Cohort

BASELINE

COHORT (MIMIC-IV)

30,615

ICU stays analysed

TEST SET

4,593

held-out patients

POSITIVE RATE

23.6%

stayed > 7 days

IMBALANCE

3.2 : 1

negative : positive

WHY ICU LENGTH OF STAY?

ICU beds are among the most expensive and scarce hospital resources. Predicting early which patients will stay longer than 7 days enables proactive resource planning — before the long stay actually unfolds.

- Early identification of high-risk patients
- Optimised bed & staff allocation
- Improved discharge planning & family communication

PREDICTION SETUP

- **Target:** binary `los_gt7` — LOS > 7 days
- **Observation window:** first 48h after ICU admission (early prediction, no outcome leakage)
- **Split:** patient-level stratified train / val / test
- **Imbalance handling:** `pos_weight` ≈ 3.2 in the loss

Finalized Architecture — Multi-Branch GRU

MODEL DESIGN

ARCHITECTURE FLOW

Input Layer

▼

Time Series	Static	Text (CXR)
48h × 13 vitals	177 features	1536d
▼	▼	▼
GRU → 64d state	**MLP** → 32d	**Linear** 1536 → 32d

▼

Fusion + Classifier Head

▼

Output: P(LOS > 7d)

BRANCHES & TRAINING

Time-series branch	**GRU over 48h × 13 vitals**
Static branch	**177 features → dense (32d)**
Text branch	**BioClinicalBERT → 32d** Sec. 2
Optimizer	**Adam (lr = 1e-3)**
Loss	**BCEWithLogitsLoss (pos_weight)**
Regularization	**Dropout · early stopping**

Class imbalance: `pos_weight` ≈ 3.2 compensates the 23.6% positive rate, prioritising recall on long-stay patients — appropriate for clinical screening.

Finalized Baseline — Test Set Results

PERFORMANCE

0.845	0.625	0.802	0.638
AUROC	F1 SCORE	ACCURACY	AUPRC

METRICS BREAKDOWN (N = 4,593)

METRIC	VALUE	NOTE
Precision	**0.564**	56% of predicted long stays correct
Recall	**0.701**	catches 70% of actual long stays
F1 Score	**0.625**	harmonic mean
AUROC	**0.845**	ranking / discrimination
AUPRC	**0.638**	key metric on imbalanced data

INTERPRETATION

Strong ranking ability (AUROC 0.845), competitive with published ICU LOS benchmarks. Recall is prioritised over precision via `pos_weight` — the right trade-off for clinical screening.

This configuration is now fixed as the baseline (full cohort, text branch enabled) against which the prototype model in Section 5 is measured.

SECTION 02

Including Textual Embeddings

Encoding free-text radiology reports with BioClinicalBERT — and comparing the full cohort against a CXR-only ablation.

02

BioClinicalBERT — Radiology Reports → Vectors

TEXT BRANCH

BioClinicalBERT (`emilyalsentzer/Bio_ClinicalBERT`) is a BERT transformer pre-trained on MIMIC-III clinical text. From a free-text section it produces a single **768-dim vector** — a semantic summary where clinically similar findings lie close together, regardless of wording.

SEMANTIC SIMILARITY EXAMPLES

"bilateral pleural effusions" ≈ "fluid accumulation in both lungs"

"no acute cardiopulmonary process" ≈ "unremarkable chest X-ray"

Inference only: BioClinicalBERT weights stay **frozen** — only the projection layer is trained. The model never "reads" text; it learns which numeric patterns in the embedding correlate with long stays.

PATIENT STATISTICS

WITH CXR (48H)	WITHOUT CXR
17.5%	82.5%
5,367 stays	25,248 stays

EMBEDDING PIPELINE

FINDINGS + IMPRESSION → **BioClinicalBERT** [CLS] · 12 layers → **Concat** 768+768 = 1536

In-model projection: `Linear(1536 → 32) → ReLU → Dropout` , then fused with the GRU state (64) and static branch (32). Only the earliest valid report (within 48h) per stay is used.

Handling Missing Reports (82.5% of Patients)

DESIGN DECISION

Patients without a CXR report receive a **zero vector (1536 × 0.0)** for the text branch. Crucially, there is **no explicit `has_cxr` flag** — an earlier version tested one and performed worse.

WHY THE FLAG HURT THE MODEL

With an explicit flag the model learns a shortcut: when `has_cxr=0` it ignores the text branch entirely. Since 82.5% of training cases have no CXR, this signal dominates training — the projection layer barely receives gradients for the 17.5% with real embeddings, so the text branch is poorly trained and contributes nothing.

WHY NO FLAG WORKS BETTER

Without a flag, the model must implicitly learn that a zero vector is uninformative. The gradient for a zero input is structurally small (`Linear(0) = 0`), so the projection layer focuses its learning capacity on the 17.5% of patients with genuine embeddings.

Implicit handling beats an explicit flag

Zero vectors let the 17.5% of patients with real reports drive the text branch — no shortcut, no dilution.

Comparison — Does the Text Branch Help?

CONFIGURATION A VS B

TEST METRICS (WINNER HIGHLIGHTED)

METRIC	A · FULL COHORT 30,615	B · CXR ONLY 5,367
Accuracy	**0.8019**	0.7935
Precision	0.5643	**0.6245**
Recall	**0.7008**	0.6752
F1 Score	0.6252	**0.6489**
AUROC	**0.8451**	0.8275
AUPRC	0.6375	**0.6539**

F1 HIGHER IN B (0.649 VS 0.625)

With a real report the model discriminates more precisely — fewer false alarms (precision 0.625 vs 0.564).

AUROC HIGHER IN A (0.845 VS 0.828)

The ×5.7 larger training set outweighs missing text — ranking ability benefits more from data volume than from complete embeddings.

TAKEAWAY

Config A is preferred for practice (covers all patients). Config B is an ablation proving the CXR embedding carries an **independent prognostic signal** on its own.

SECTION 03

Finalizing Explainability

SHAP and TimeSHAP reveal which features — and which ICU hours — drive each prediction, and surface three open data-quality issues.

03

Method Choice — Why SHAP, Not LIME

METHODOLOGY

CRITERION	LIME	SHAP + TIMESHAP
Approach	Local linear fit	Shapley values (game theory)
Stability	Unstable	Consistent
Scope	Local only	Global + local
Time series	Steps independent	Each of 48 hours = 1 player

Decision: LIME excluded. Instability is unacceptable in a clinical context where reproducibility is required for trust and regulatory compliance.

IMPLEMENTATION

- `shap.GradientExplainer` (backprop gradients)
- Background: 200 random training patients as the "average patient"
- TimeSHAP recovers the *when* that standard SHAP collapses away

METHODOLOGY SOURCE

Hsieh et al. (2024). *A Comprehensive Guide to Explainable AI: From Classical Models to LLMs.* arXiv:2412.00800v2, 255 pp. — SHAP pp. 71–73 · LIME instability p. 76 · TimeSHAP for RNNs pp. 116–118.

SHAP — Which Features Drive Predictions

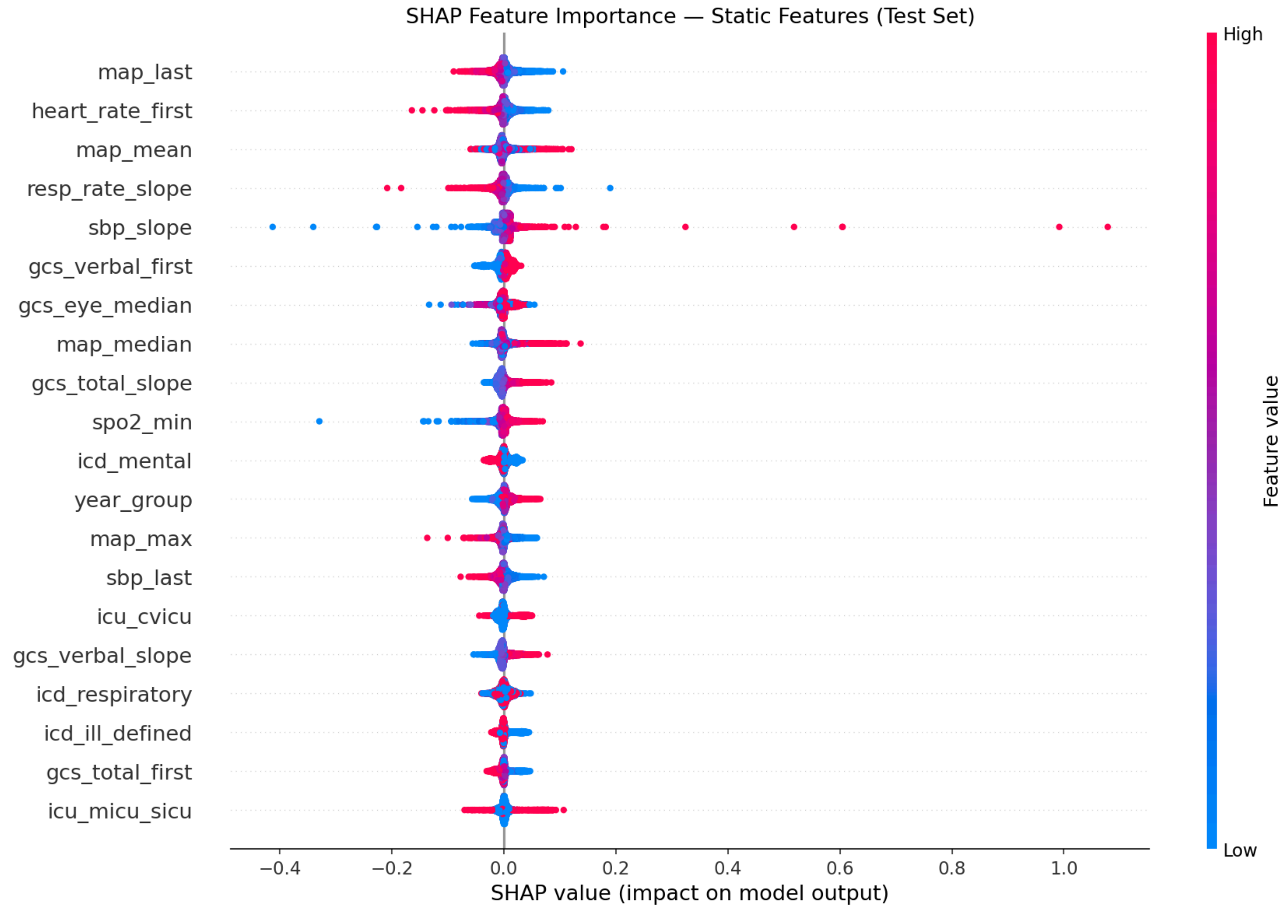


Fig. 1 — SHAP beeswarm. Each dot = one patient · red = high feature value, blue = low.

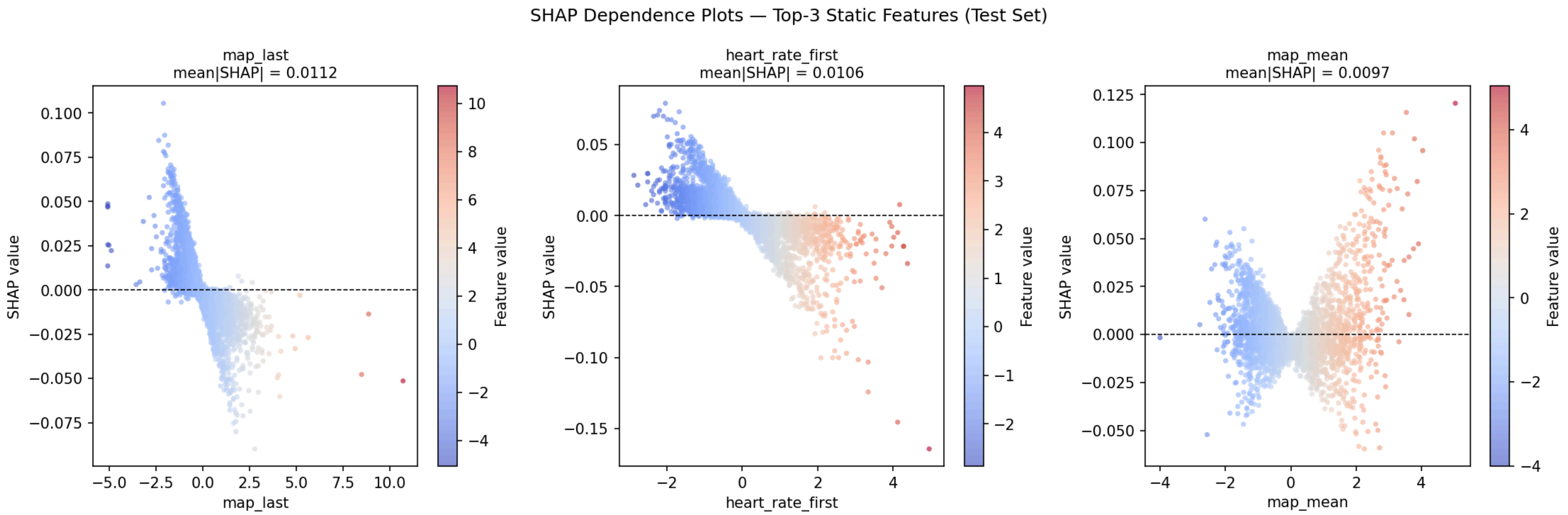
TOP-5 FEATURES

#	FEATURE	MEAN|SHAP|
1	`map_last`	0.0112
2	`heart_rate_first`	0.0106
3	`map_mean`	0.0097
4	`resp_rate_slope`	0.0090
5	`gcs_verbal_first`	0.0088

Assessment: MAP (blood pressure), heart rate at admission, and GCS (consciousness) are established ICU severity indicators. The model learned **real medical patterns** — not statistical artefacts.

SHAP Dependence — Physiological Thresholds

TOP-3 FEATURES



MAP_LAST

Low MAP (hypotension) increases predicted risk.

HEART_RATE_FIRST

High heart rate at admission increases risk.

MAP_MEAN

Low average BP over 48h correlates with longer stay.

Fig. 2 — non-linear patterns confirm the GRU learned physiological thresholds

SHAP Waterfall — A False Positive Dissected

PER-PATIENT

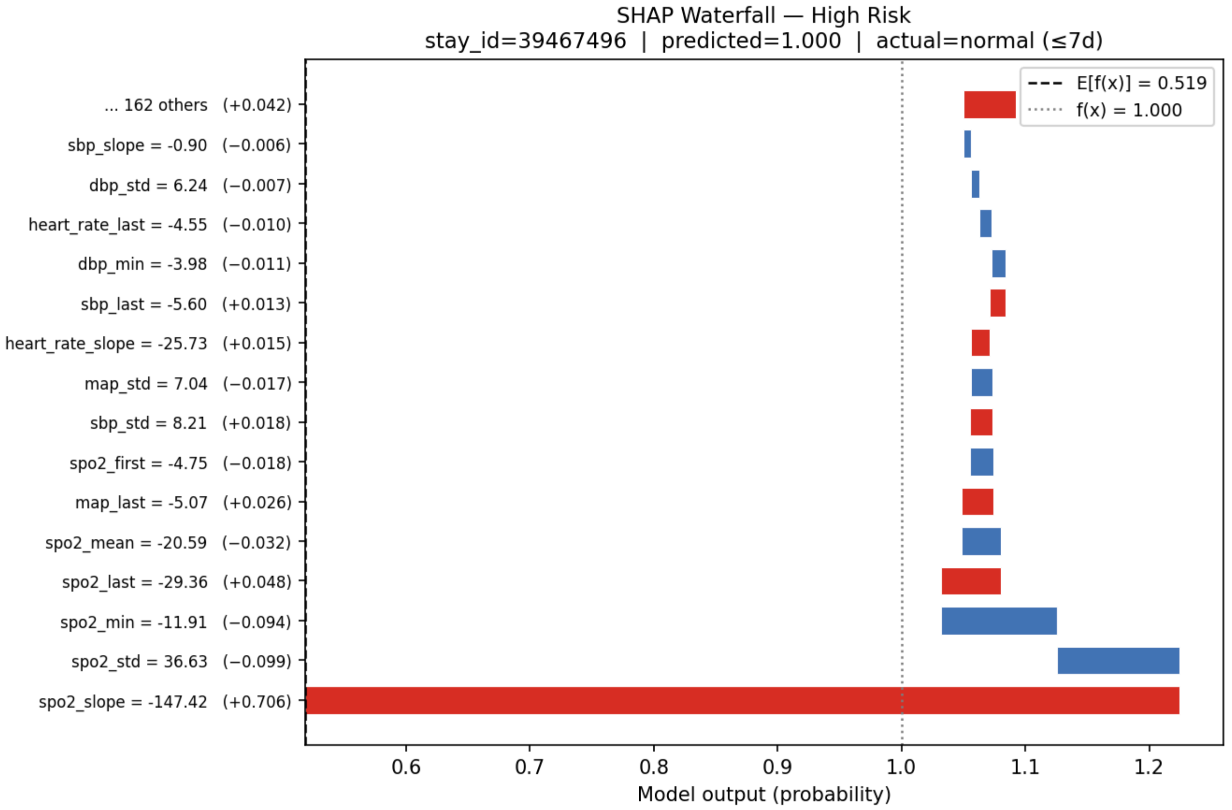


Fig. 3 — stay_id 39467496: predicted 1.000, actual = normal stay (≤7d). Baseline E[f(x)] = 0.519.

PROBLEM IDENTIFIED

`spo2_slope = −147` (a 48h SpO_2 trend) contributed SHAP **+0.706**, pushing the prediction to its maximum. Root cause: a likely **sensor artefact** — the model cannot tell real deterioration from data noise.

RECOMMENDED FIX

Winsorize aggregated vital features (clip to p1–p99 of the training set, fit on train only). This would clip `spo2_slope = −147` to ≈ −5. **Not yet implemented**

TimeSHAP — At Which ICU Hour Did It Matter?

TEMPORAL ATTRIBUTION

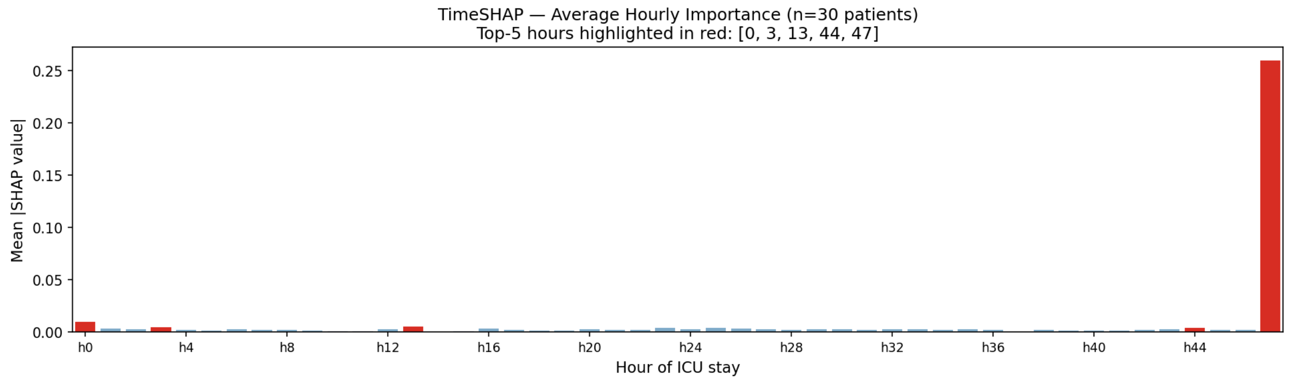


Fig. 4 — Average hourly TimeSHAP importance (30 patients). Top-5: h0 0.010 · h3 0.005 · h13 0.005 · h44 0.004 · h47 0.260.

H0 — CLINICALLY CORRECT

The admission hour is the most important — the initial patient state drives LOS.

H47 — AN ARTEFACT

ffill propagates the last measurement into h47, so zeroing it removes the most recent state → largest SHAP. Background = zeros (not mean), since ffill/bfill makes patient values ≈ mean. Fix: explicit missingness flags.

Not implemented

Explainability — What Works & What's Open

OUTCOME

✓ WHAT WORKS

- SHAP identifies MAP, HR, GCS as top features — clinically plausible
- Dependence plots confirm correct physiology (hypotension = higher risk)
- The GRU learned non-linear clinical patterns
- TimeSHAP correctly flags admission hour h0 as most important
- Method choice backed by Hsieh et al. (2024)

⚠ OPEN ISSUES (NOT YET IMPLEMENTED)

ISSUE	ROOT CAUSE	RECOMMENDED FIX
h47 dominance	ffill/bfill creates correlated hours	Per-hour missingness flags
FP spo2_slope = −147	Sensor artefact not filtered	Winsorize to p1–p99 (train)
ECE = 0.29 (overconfident)	Probabilities not calibrated	Post-hoc Platt scaling

SECTION 04

Time-Series Analysis (Frequency of Values)

How frequently each ICU variable is actually measured in the first 48 hours — and what that irregularity means for modelling.

04

Objective & Methodology

Characterize how ICU variables are monitored during the first 48h after admission, across **~30,000 ICU stays** using Chartevents + Outputevents.

METRICS

- **Coverage** — proportion of ICU hours observed (Observed / 48)
- **Mean Gap** — average interval between measurements
- **Max Gap** — longest monitoring interruption

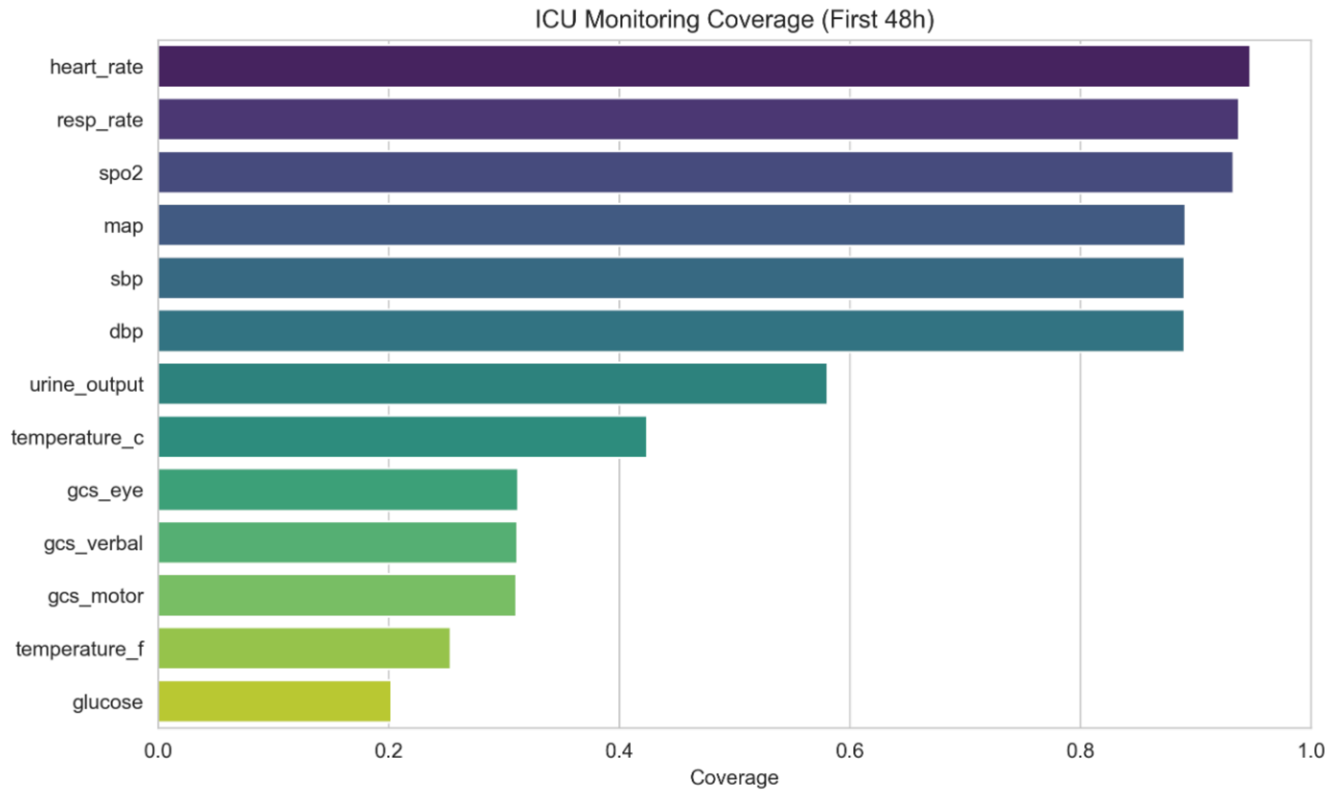
MONITORING REGIMES

REGIME	INTERVAL
Continuous	≤ 1.5 h
Frequent	1.5 – 4 h
Intermittent	4 – 12 h
Sparse	12 – 24 h
Episodic	> 24 h

Goal: quantify temporal monitoring intensity across clinical variables.

Coverage of ICU Variables (First 48h)

COVERAGE



Coverage = observed ICU hours / 48, per clinical variable.

HIGH (> 80%)

Vital signs (HR, RR, SpO_2, BP) cover nearly the entire stay.

MODERATE (40–60%)

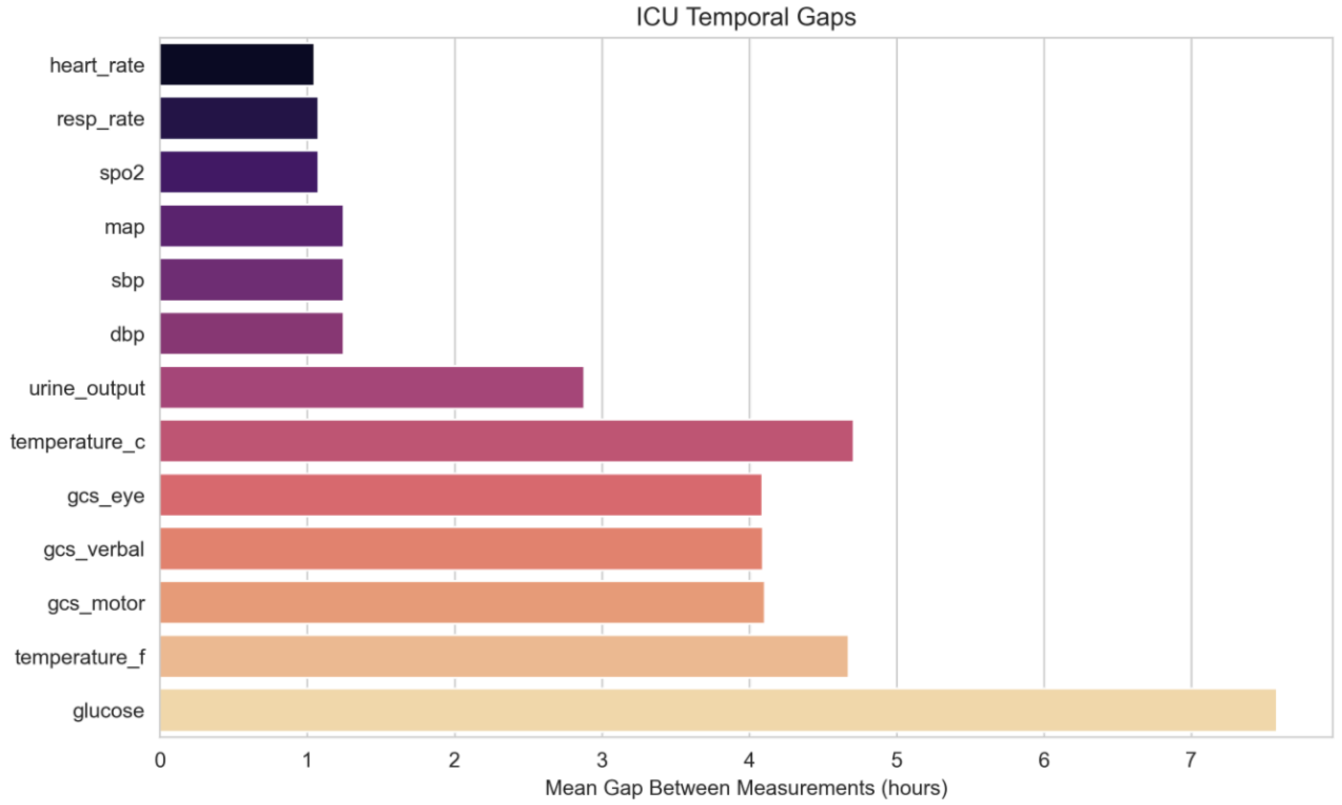
Urine output & temperature are moderately monitored.

LIMITED (< 25%)

Neurological assessments (GCS) and glucose have limited temporal coverage.

Average Time Between Measurements

FREQUENCY



Mean gap between measurements (hours), per variable.

- **Vital signs:** measured roughly every hour
- **Urine output:** documented every ~3 hours
- **GCS & temperature:** assessed every ~4–5 hours
- **Glucose:** measured most sparsely (> 7 h)

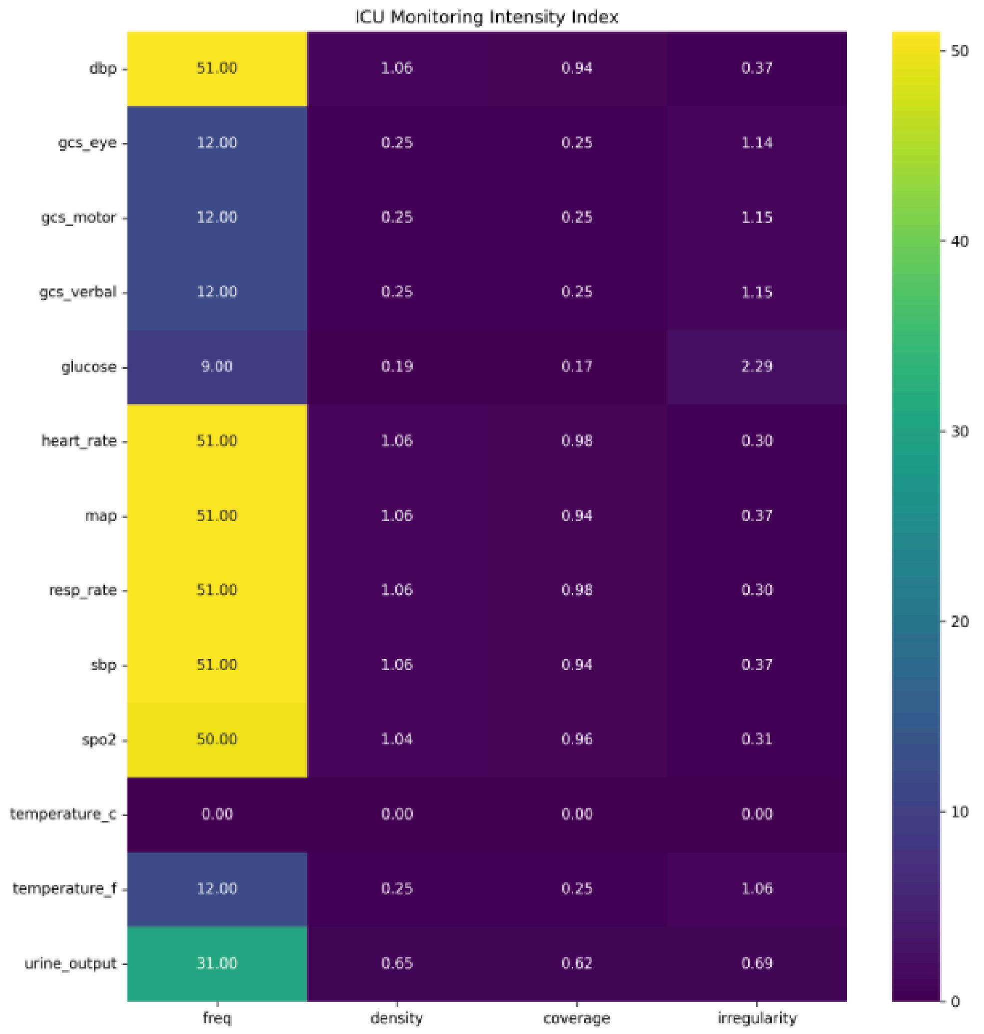
Implication: ICU time series are inherently irregular — uniform-sampling assumptions are inappropriate, and monitoring intensity itself may be clinically meaningful.

Continuous vs Intermittent & Intensity Index

REGIMES

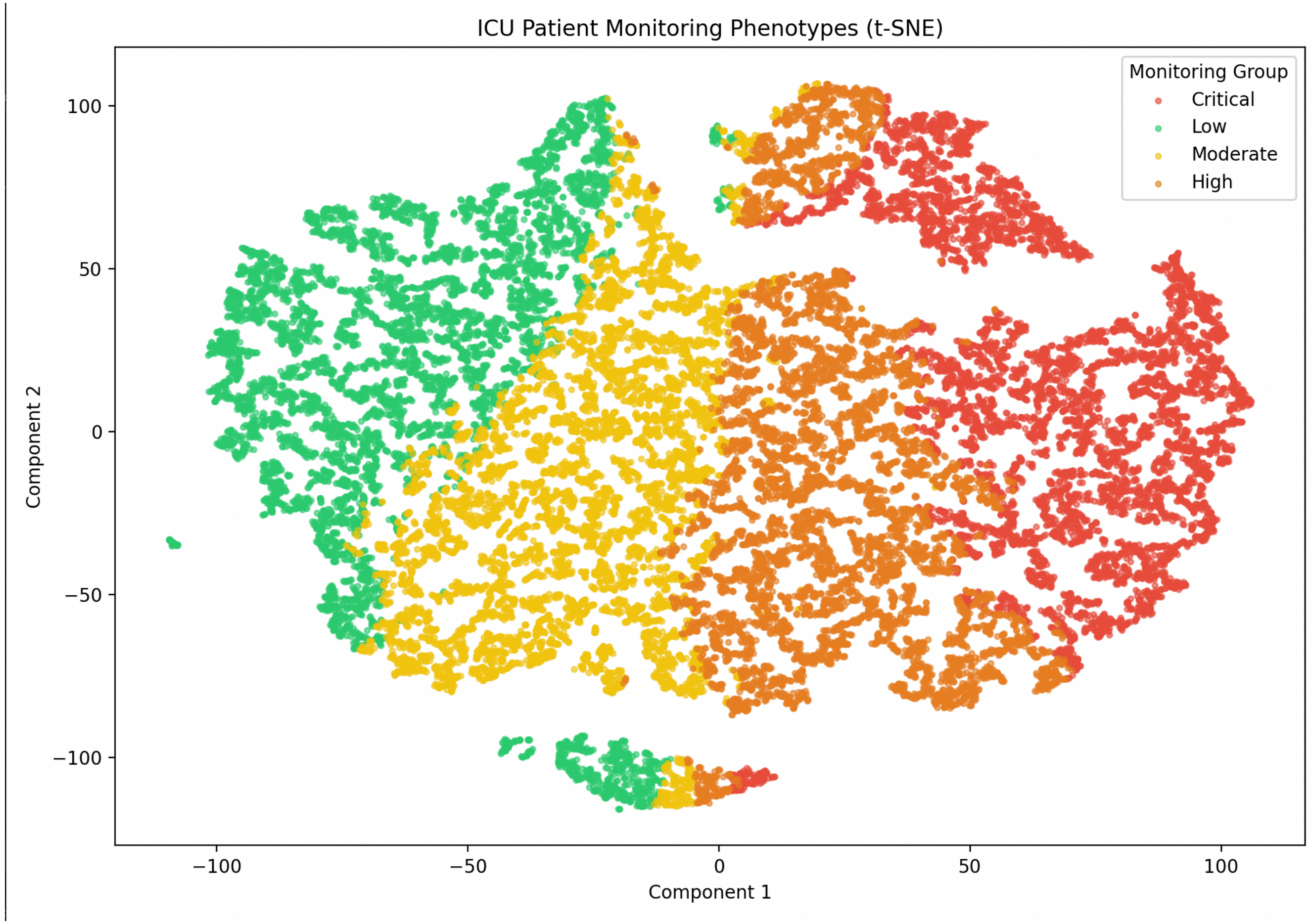
CONTINUOUS-MONITORING RATE

Heart Rate	**98.5%**
Respiratory Rate	97.8%
SpO_2	97.6%
Blood Pressure	92.8%
Temperature	74.8%
Urine Output	50.7%
GCS	**9.0%**
Glucose	**3.2%**



ICU Monitoring Intensity Index: freq / density / coverage / irregularity per variable.

Patient Phenotypes — Clustering (t-SNE)



t-SNE of the Patient Monitoring Intensity (PMI) features — four phenotype groups.

PMI SCORE

$$\mathrm{PMI} = 0.4 \cdot Z_{freq} + 0.2 \cdot Z_{coverage} + 0.2 \cdot Z_{maxfreq} + 0.2 \cdot Z_{features}$$

Low Moderate High Critical

The clusters form a **continuous gradient** of monitoring intensity — confirming that measurement availability is not random but reflects patient condition and clinical workflow.

Conclusions & Implications for Modelling

FINDINGS

① DISTINCT REGIMES

Continuous (HR, RR, SpO_2, BP) · Periodic (Temp, Urine) · Intermittent (GCS, Glucose). ICU variables are monitored very differently.

② MISSINGNESS ≠ RANDOM

Availability reflects clinical workflow, monitoring technology, and patient condition — not random missingness.

③ MODELLING IMPACT

Time series are inherently irregular; uniform-sampling assumptions are inappropriate; monitoring intensity itself is informative.

Connection: this directly motivates the open explainability fixes (missingness flags for the h47 artefact) and the prototype model's reliance on learned embeddings rather than raw uniform time steps.

SECTION 05

First Prototype Model Implementation

PRD-Net: a peer-retrieval derivation network — and how its results compare to the finalized baseline.

05

PRD-Net — Peer Retrieval Derivation Network

CONCEPT

Core idea: instead of classifying directly from the embedding, PRD-Net asks — *how similar is this patient to clinically comparable patients who stayed long vs. short?*

Step 1 GRU Embeddings (128d) → **Step 2** Peer Groups K=20 → **Step 3** MLP 128→64 + head → **Step 4** Train (BCE + pos_weight) → **Step 5** Inference

DISTANCE-WEIGHTED

Prototypes = inverse-distance weighted mean of the 20 nearest peers per class — nearer peers count more.

CLINICALLY FILTERED

Same ICD + ICU + age filter applied end-to-end (train, val, test) for a consistent prototype structure.

INTERPRETABLE

Every prediction is grounded in concrete, retrievable peer patients — intrinsic interpretability.

Filter Experiments & Class Imbalance

TUNING

PEER-FILTER ABLATIONS

HARD / SOFT FILTER	K	ACC	F1
ICU + ICD · Age ±10	20	0.800	0.586
+ Admission type	20	0.804	0.602
+ GCS ±3	20	0.806	0.586
+ GCS ±3	15	0.802	0.576

Extra filters shrink the peer pool and force fallbacks. **ICD + ICU + age already captures the relevant structure** — the GRU embeddings handle the rest.

POS_WEIGHT IN THE LOSS

With 23.7% long-stay (ratio ≈ 3.23:1), a naive model hits ~76% accuracy by always predicting short. Setting `pos_weight = 16560/5130 ≈ 3.23` makes each false negative weigh 3.23× a false positive.

+3.8pp
F1: 0.586 → 0.624

–2.0pp
ACC: 0.800 → 0.780

Final config: ICU + ICD + Age ±10, K=20, pos_weight 3.23 → F1 0.624 · AUROC 0.839 · AUPRC 0.627.

Results — Prototype vs Baseline vs Literature

COMPARISON

VS OUR BASELINE (SECTION 1)

METRIC	BASELINE GRU	PRD-NET
F1 Score	0.625	0.624
AUROC	0.845	0.839
AUPRC	0.638	0.627

VS PUBLISHED MODELS (WU ET AL. 2021)

MODEL	AUROC	AUPRC
Wu — SAPS II	0.670	0.462
Wu — Random Forest	0.737	0.520
Wu — GBDT (best)	0.747	0.536
PRD-Net (ours)	**0.839**	**0.627**

Verdict: PRD-Net **matches the baseline** within ~1pp on every metric while adding intrinsic peer-based interpretability — and beats the best Wu model by ≈ 9pp AUPRC using structured features alone.

COMPARISON SOURCE (OUR REFERENCE)

Wu, J. et al. (2021). *Predicting Prolonged Length of ICU Stay through Machine Learning.* Diagnostics 11(12), 2242. pmc.ncbi.nlm.nih.gov/articles/PMC8700580 — RF / SVM / DL / GBDT for pLOS-ICU on eICU-CRD, externally validated on MIMIC-III.

AUPRC is the key metric on imbalanced data (ignores true negatives)

PRD-Net — Next Steps

ROADMAP

Medium effort

FOCAL LOSS

Replace weighted BCE with Focal Loss — penalises hard, misclassified cases more. Targets borderline patients near the 7-day threshold.

High effort · highest impact

END-TO-END TRAINING

Embeddings are currently frozen. Fine-tuning the encoder jointly lets it learn representations optimised for peer retrieval — architecturally the largest lever.

High effort

TEXT EMBEDDINGS

Incorporate clinical free-text notes as an additional modality — connecting back to the BioClinicalBERT work in Section 2.

End-to-end training is the single biggest lever

Unfreezing the GRU encoder lets representations be optimised directly for the peer-retrieval task — the largest expected gain.

Summary — Where We Stand

WRAP-UP

① BASELINE FINALIZED

GRU multi-branch model: AUROC 0.845, F1 0.625 on 4,593 test patients.

② TEXT ADDS SIGNAL

BioClinicalBERT CXR embeddings carry an independent prognostic signal (ablation B).

③ EXPLAINABLE

SHAP + TimeSHAP validate clinical patterns; 3 data-quality fixes identified.

④ IRREGULAR BY DESIGN

Monitoring analysis shows non-random missingness across variable types.

⑤ PROTOTYPE MATCHES BASELINE

PRD-Net reaches AUROC 0.839 with intrinsic peer-based interpretability.

NEXT

End-to-end training, calibration (Platt), winsorization, and text fusion into the prototype.

THANK YOU

Questions & Discussion

ICU Length-of-Stay Prediction · MIMIC-IV

Comparison model: Wu et al. (2021). Predicting Prolonged Length of ICU Stay through Machine Learning. *Diagnostics* 11(12), 2242. https://pmc.ncbi.nlm.nih.gov/articles/PMC8700580/

Explainability methodology: Hsieh et al. (2024). A Comprehensive Guide to Explainable AI. arXiv:2412.00800v2.

Data & models: MIMIC-IV / MIMIC-CXR (PhysioNet) · BioClinicalBERT (emilyalsentzer/Bio_ClinicalBERT).